

NGO IMPACT MEASUREMENT DASHBOARD

Transforming Raw Field Data into Actionable Impact Insights



CHALLENGE THEME

Impact Measurement Dashboard
for a Partner NGO



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THE PROBLEM

Why Impact Measurement Matters for NGOs



CURRENT CHALLENGE

NGOs track activities, but struggle to measure real impact



Beneficiaries
Reached



Programs
Conducted



Funds
Utilized



Attendance
Tracked

KEY QUESTIONS

- Did beneficiaries actually improve?
- Which programs create the greatest impact?
- Are resources being used efficiently?
- What outcomes are being achieved?

BUSINESS PROBLEM



Reactive
Decisions



High-impact
Programs
Unidentified



Limited
Evidence for
Donors



Inefficient
Resource
Allocation

THE GAP



Students
Enrolled



Programs
Delivered



Data
Collected



Impact
Unknown



TO

OUR SOLUTION



Raw Field
Data



Smart
Dashboard



Actionable
Insights



Measurable
Impact



Opportunity: Transform Data into
Decisions that Drive Real Impact



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PROJECT OBJECTIVES & SOLUTION OVERVIEW

From Raw Data to Measurable Impact



PROJECT GOAL

Design and develop an interactive NGO Impact Measurement Dashboard that transforms raw field data into actionable insights for decision-makers.

KEY OBJECTIVES



01 MEASURE PROGRAM EFFECTIVENESS

- Track beneficiary learning outcomes
- Monitor improvement across programs
- Identify high-performing interventions



02 MONITOR BENEFICIARY REACH

- Analyze participation across districts
- Understand demographic coverage
- Track program completion



03 EVALUATE IMPACT

- Measure outcome achievement
- Assess learning improvement
- Connect activities to measurable impact



04 IMPROVE COST EFFICIENCY

- Calculate cost per successful beneficiary
- Support data-driven resource allocation
- Enhance donor reporting



DELIVERABLES

- ✓ Impact Measurement Dashboard
- ✓ Data Dictionary
- ✓ User Guide for NGO Staff



VALUE CREATED



Activity Tracking



Outcome Measurement



Impact Evaluation



Better Decisions

OUR SOLUTION APPROACH



RAW NGO DATA

Data collected from field operations



DATA CLEANING & VALIDATION

Handling missing values, duplicates and ensuring data quality



FEATURE ENGINEERING

Creating key metrics like improvement %, attendance bins, completion status, outcome achievement



KPI FRAMEWORK

Activity → Output → Outcome → Impact
Key performance indicators defined



INTERACTIVE DASHBOARD

Visualizing insights through charts, KPIs and filters for easy analysis



ACTIONABLE DECISION MAKING

Data-driven decisions to improve programs, allocate resources and maximize impact



MEASURABLE
SOCIAL IMPACT

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DATASET OVERVIEW & DATA PREPARATION

Building a Reliable Foundation for Impact Measurement



OUR DATA JOURNEY



300 RECORDS
Raw Beneficiary Data



DATA CLEANING
Handling Missing Values & Inconsistencies



DATA VALIDATION
Applying Business Rules & Quality Checks



FEATURE ENGINEERING
Creating Metrics & Derived Features



READY FOR ANALYSIS
Clean, Reliable, and Analysis-Ready Data



DATASET OVERVIEW

The project uses a beneficiary-level dataset representing NGO educational programs across multiple districts.

METRIC	VALUE
 Total Beneficiaries	300
 Districts Covered	5
 Programs	4
 Gender Categories	Male, Female
 Records Analyzed	300



DATA QUALITY CHALLENGES

Before analysis, the dataset contained:

- Missing Gender values
- Missing District values
- Missing Post-Test Scores

These issues could impact KPI accuracy and dashboard reliability.



DATA CLEANING PROCESS

Missing Value Treatment

VARIABLE	METHOD
Gender	Mode Imputation
District	Mode Imputation
Post-Test Score	Median Imputation

KEY VARIABLES



Beneficiary Information

- Student ID
- Age
- Gender
- District



Program Information

- Program Name
- Program Cost
- Attendance Percentage



Learning Outcome Information

- Pre-Test Score
- Completion Status
- Post-Test Score
- Outcome Achieved

OUTCOME



Raw Data



Cleaned Dataset



Reliable KPIs



Accurate Decision-Making



High-quality data leads to trustworthy insights and meaningful impact.

EXPLORATORY DATA ANALYSIS & FEATURE ENGINEERING

Transforming Data into Actionable Metrics



WHY EDA WAS IMPORTANT

Before building the dashboard, exploratory data analysis (EDA) was performed to:

- ✓ Understand beneficiary behavior
- ✓ Identify learning trends
- ✓ Measure program effectiveness
- ✓ Discover factors influencing outcomes



KEY QUESTIONS EXPLORED

- Does attendance influence learning improvement?
- Which program produces the highest improvement?
- Do higher attendance groups achieve better outcomes?
- What is the relationship between cost and impact?



KEY METRICS CREATED

To enable impact measurement, new features were engineered from the raw dataset.



1. IMPROVEMENT PERCENTAGE

$$\frac{(\text{Post-Test Score} - \text{Pre-Test Score})}{\text{Pre-Test Score}} \times 100$$

Purpose:

- Quantify learning gains
- Compare program effectiveness



2. ATTENDANCE BINS

25-50%

50-75%

75-100%

Purpose:

- Analyze attendance impact on outcomes
- Simplify trend analysis



3. COMPLETION STATUS

Yes / No

Purpose:

- Track program completion rates



4. OUTCOME ACHIEVED

Yes / No

Purpose:

- Measure whether beneficiaries met target outcomes



OUTCOME

The EDA phase revealed strong relationships between:



Attendance and Improvement



Program Type and Performance



Attendance and Outcome Achievement

These findings directly informed dashboard design and KPI selection.

ANALYTICS WORKFLOW



1. RAW DATASET

Collected from field operations



2. DATA CLEANING

Handled missing values & inconsistencies



3. EXPLORATORY DATA ANALYSIS (EDA)

Patterns, trends & relationships discovered



4. FEATURE ENGINEERING

Created new metrics & categories



5. INSIGHTS GENERATION

Key insights & drivers identified



6. DASHBOARD DEVELOPMENT

Converted insights into interactive dashboard

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DASHBOARD DESIGN & KPI FRAMEWORK

Converting Data into Actionable Insights



NGO IMPACT MEASUREMENT DASHBOARD

Gender

All

District

All

Program Name

All



TOTAL STUDENTS
REACHED

300



TOTAL PROGRAM
COST

₹ 2,11,000



COMPLETION
RATE

95.33%



OUTCOME
ACHIEVEMENT RATE

60.33%



AVERAGE
IMPROVEMENT (%)

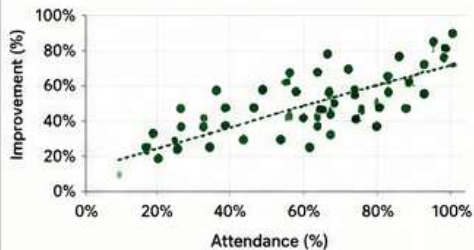
35.30%



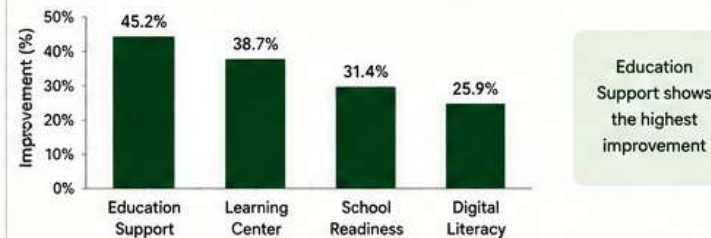
COST PER
SUCCESSFUL STUDENT

₹ 1,165

ATTENDANCE vs LEARNING IMPROVEMENT



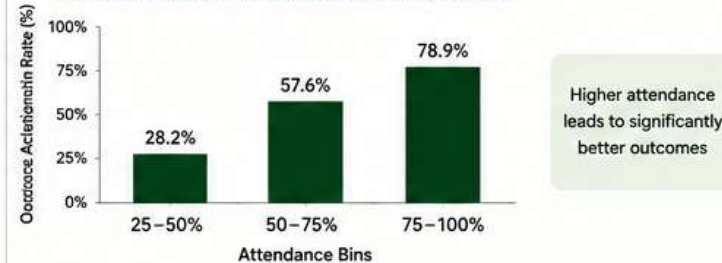
AVERAGE IMPROVEMENT BY PROGRAM



OUTCOME ACHIEVEMENT DISTRIBUTION



ATTENDANCE BIN vs OUTCOME ACHIEVEMENT



1 KPI CARDS – 6 CORE METRICS

At-a-glance view of key performance indicators



Students
Reached



Program
Cost



Completion
Rate



Outcome
Achievement
Rate



Avg.
Improvement
(%)



Cost per
Successful
Student

2 INTERACTIVE FILTERS

Drill down and analyze data by



Gender



District



Program Name

3 VISUAL ANALYTICS – 4 KEY CHARTS



- Attendance vs Learning Improvement
- Average Improvement by Program
- Outcome Achievement Distribution
- Attendance Bin vs Outcome Achievement

4 KPI IMPACT FRAMEWORK

Aligning activities to long-term social impact



Activity
Student Enrollment
Program Delivery



Output
Program Completion
Beneficiaries Reached



Outcome
Learning Improvement
Outcome Achievement



Impact
Cost-Effective Success
Long-Term Impact

DASHBOARD VALUE – FROM DATA TO IMPACT



Raw Data
(Multiple Sources)



Meaningful KPIs
& Metrics



Actionable
Insights



Better
Decisions



Greater
Impact

THE ONE NUMBER TO CHECK EVERY MONDAY

THE OUTCOME ACHIEVEMENT RATE



60.33%

- ✓ Directly measures program effectiveness
- ✓ Focuses on beneficiary outcomes
- ✓ Clear indicator of real impact

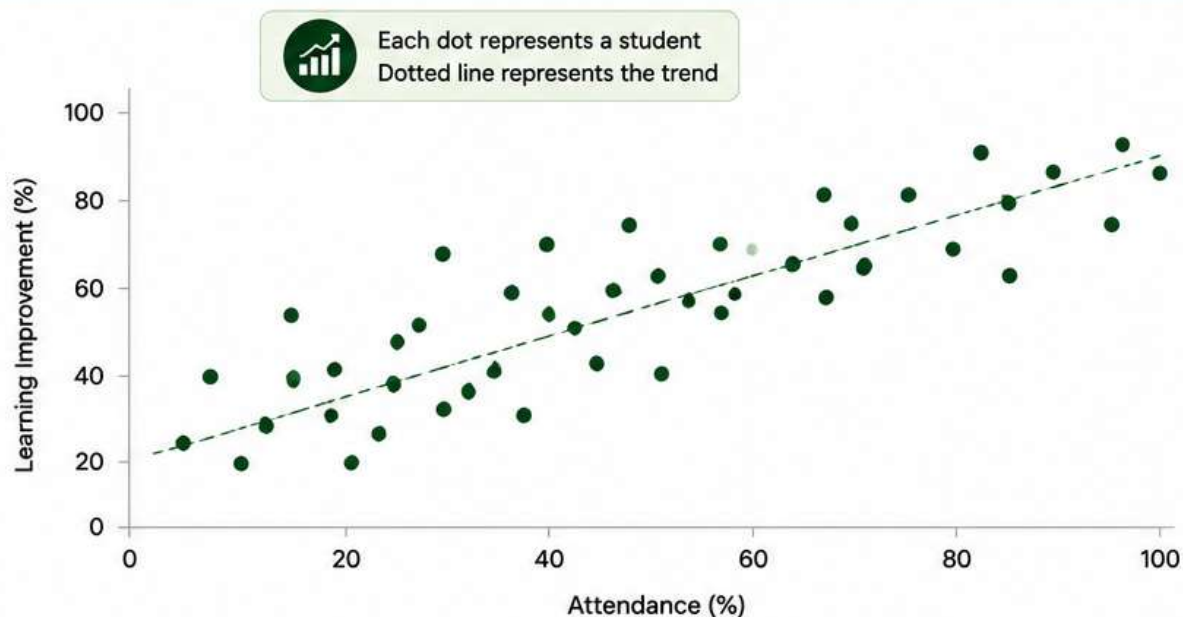
This KPI tells us how many beneficiaries are actually achieving the intended outcomes.

07 KEY INSIGHT #1: ATTENDANCE DRIVES LEARNING IMPROVEMENT

Higher Attendance Leads to Better Learning Outcomes



ATTENDANCE VS LEARNING IMPROVEMENT



The trend clearly shows that as **attendance** increases, **learning improvement** also increases.



KEY FINDING

A strong positive relationship exists between attendance and learning improvement.

Students with higher attendance consistently achieved greater score improvements.



EVIDENCE FROM THE DATA

- Improvement percentage increases as attendance increases.
- Students attending 75–100% of sessions demonstrate significantly better performance.
- The trend line confirms a positive correlation between attendance and learning gains.



WHY THIS MATTERS

- ✓ Better learning outcomes
- ✓ Greater program effectiveness
- ✓ Higher outcome achievement rates
- ✓ Improved return on program investment



RECOMMENDATION

The NGO should prioritize attendance improvement initiatives such as:

- SMS/WhatsApp reminders
- Incentive-based participation
- Parent engagement programs
- Community outreach campaigns



BUSINESS QUESTION

Can improving attendance improve beneficiary outcomes?

IMPACT PATH



Higher Attendance



Better Learning



Higher Outcome Achievement



Greater Social Impact

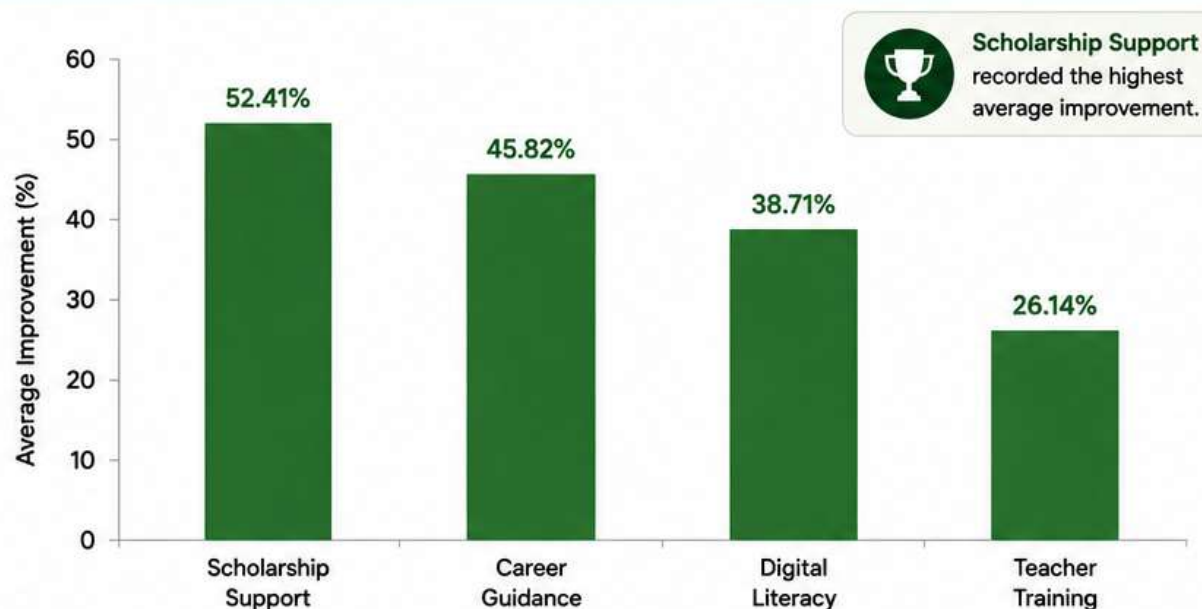
08

KEY INSIGHT #2: PROGRAM PERFORMANCE COMPARISON

Which NGO Program Creates the Greatest Impact?



AVERAGE IMPROVEMENT (%) BY PROGRAM



PROGRAM RANKING



BUSINESS QUESTION

Which program delivers the highest learning improvement and should be prioritized for future investment?

IMPACT PATH



KEY FINDING

Scholarship Support achieved the highest average improvement among all programs.
Not all programs generate the same level of impact.



WHAT THE DATA TELLS US

- Not all programs generate the same level of impact.
- Scholarship Support consistently produced stronger learning gains.
- Teacher Training showed comparatively lower improvement and may require further evaluation.



WHY THIS MATTERS

- ✓ Allocate resources more effectively
- ✓ Scale successful interventions
- ✓ Improve underperforming programs
- ✓ Maximize social impact per rupee spent



RECOMMENDATION

SHORT-TERM

- Continue investing in Scholarship Support initiatives.
- Document best practices from high-performing programs.

LONG-TERM

- Apply successful strategies from Scholarship Support to other programs.
- Reassess delivery methods for Teacher Training programs.



Data reveals which programs create impact — allowing NGOs to invest where results are strongest.

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KEY INSIGHT #3: OUTCOME ACHIEVEMENT & COST EFFICIENCY

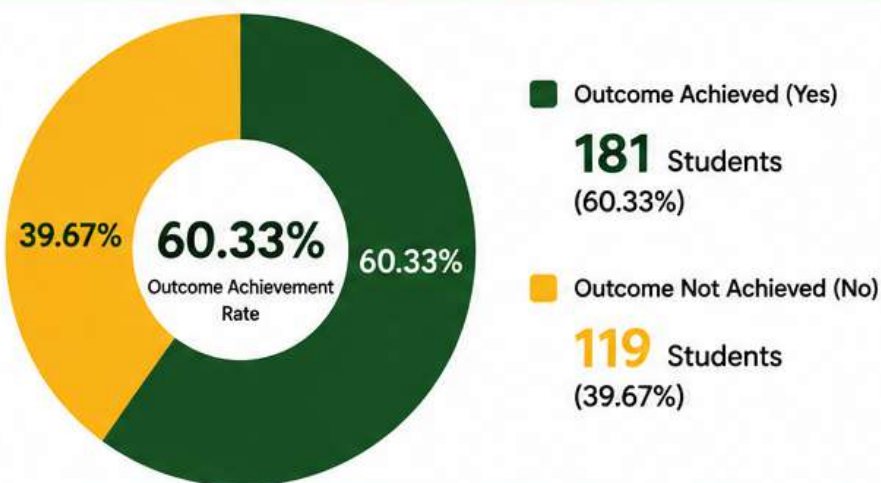
Measuring Real Impact Beyond Activities



BUSINESS QUESTION

Did the NGO successfully achieve its intended outcomes, and how efficiently were resources utilized?

OUTCOME ACHIEVEMENT DISTRIBUTION



ANALYSIS

To evaluate impact, we analyzed Outcome Achievement Rate, Average Learning Improvement, and Cost per Successful Student.

KEY FINDINGS



60.33%

OUTCOME ACHIEVEMENT RATE

181 out of 300 beneficiaries achieved the desired outcome.



30.95%

AVERAGE IMPROVEMENT

Beneficiaries improved their assessment scores by nearly 31% on average.



₹2.78K

COST PER SUCCESSFUL STUDENT

The NGO spends approximately ₹2,780 for every successful beneficiary.



WHAT THIS MEANS

The NGO is not only reaching beneficiaries but also generating measurable learning outcomes.

The dashboard shifts the focus from:



STUDENTS
REACHED



STUDENTS
SUCCESSFULLY
IMPACTED



WHY THIS MATTERS

- ✓ Better donor reporting
- ✓ Improved funding justification
- ✓ Stronger accountability
- ✓ Evidence-based decision making



EXECUTIVE KPI

OUTCOME ACHIEVEMENT RATE

60.33%

Why is this the most important KPI?

- ✓ Measures actual program success
- ✓ Reflects beneficiary impact
- ✓ Aligns with NGO mission and goals
- ✓ Provides a single indicator for leadership review



RECOMMENDATION

MAINTAIN

- Current high-performing interventions
- Outcome-focused monitoring

IMPROVE

- Increase completion rates
- Strengthen attendance initiatives
- Support underperforming beneficiary groups

IMPACT FRAMEWORK VALIDATION



ACTIVITY
Program
Activities



OUTPUT
Program
Delivery



OUTCOME
Learning
Improvement



IMPACT
60.33%
Beneficiaries
Achieved Outcomes



Impact is not measured by the number of beneficiaries reached, but by the number of beneficiaries whose lives improved.

FUTURE IMPACT & RECOMMENDATIONS

From Data to Sustainable Social Impact



PROJECT OUTCOMES

Through this dashboard, the NGO can now:

- ✓ Measure program effectiveness
- ✓ Track beneficiary outcomes
- ✓ Monitor cost efficiency
- ✓ Support evidence-based decision making
- ✓ Demonstrate impact to donors and stakeholders



STRATEGIC RECOMMENDATIONS

1 IMPROVE ATTENDANCE RATES

- Implement reminder campaigns
- Strengthen beneficiary engagement
- Increase community participation

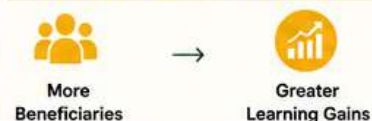
EXPECTED RESULT



2 SCALE HIGH-PERFORMING PROGRAMS

- Expand Scholarship Support initiatives
- Replicate successful delivery models
- Share best practices across programs

EXPECTED RESULT



3 STRENGTHEN OUTCOME MONITORING

- Track outcome achievement regularly
- Review KPIs monthly
- Use dashboard insights for program reviews

EXPECTED RESULT



FUTURE ENHANCEMENTS

SHORT-TERM

- Automated data refresh
- Mobile-friendly dashboard
- Real-time beneficiary tracking

LONG-TERM

- Predictive analytics for at-risk students
- Donor reporting automation
- AI-powered impact forecasting
- Multi-NGO deployment model



POTENTIAL ORGANIZATIONAL IMPACT

AREA	EXPECTED BENEFIT
Program Management	Faster decision making
Monitoring & Evaluation	Improved outcome tracking
Donor Reporting	Greater transparency
Resource Allocation	Better investment decisions
Beneficiaries	Improved learning outcomes



FINAL TAKEAWAY



“

Real impact is not measured by how many people are reached, but by how many lives are improved.

”

TEAM

Harsh Rishi

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